

1 Conditioning c

FIMr1p1 ensemble guidance

PR + T2M weekly summaries:
mean, spread, q10, q90, count

Observed predictors

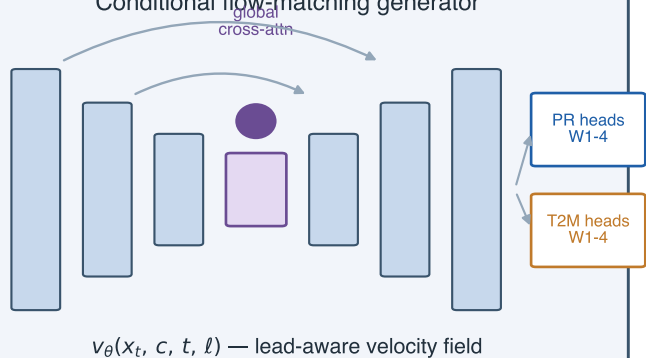
SST, SSS, soil moisture, IVT,
Z500*, U250, MJO (4 prior weeks)

Static + calendar

elevation, land mask, lat/lon
encodings, season, lead week

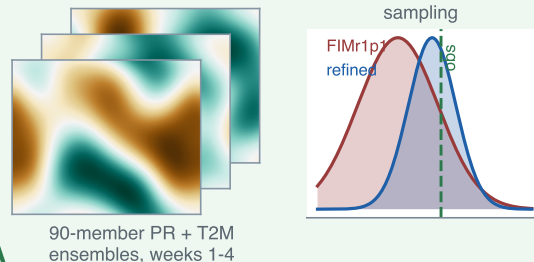
2

Conditional flow-matching generator



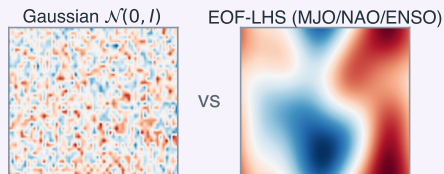
4

Weekly probabilistic forecasts



3

Physically informed stochastic prior x_0

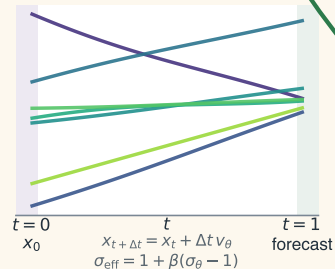


$$x_0 = \rho x_{\text{EOF}} + \sqrt{1 - \rho^2} x_{\text{rand}}$$

$\rho_{\text{PR}} = 0.25, \rho_{\text{T2M}} = 0.08$
regime- and lead-conditioned



ODE sampling (50 Euler steps)



Verification

CRPS

RMSE / bias
/ corr

calibrated
BSS

spread skill

season x lead

event tails
(q95/q99)